Lecture 1

The qubit

Objectives. *A quantum system is given by its* states*, its* observables*, a* Hamiltonian *that moves the states, and a rule for the* probabilities *of readings and the state after a reading. This lecture builds all four for the simplest system, a quantum bit, starting from silver atoms in a magnet. The states are the points of a sphere, and the Hamiltonian turns the sphere.*

1.1 Silver atoms and two magnets

Silver atoms from an oven pass through a Stern–Gerlach magnet, whose field is strong at one pole and weak at the other, and land on a screen. The uneven field pushes on the atom's *magnetic moment*, the vector that measures how the atom responds to a magnetic field. The screen shows two separated beams (Sakurai and Napolitano, Figs 1.1–1.2). They correspond to the two readings of the atom's intrinsic angular momentum, its *spin*, along the magnet's direction. This lecture is the theory of the spin alone. We record the two readings as +1 and −1.

Keep only the atoms that read +1 in a magnet along z. Between the magnets the atoms fly through no field. Three facts about what happens next are what this lecture must explain (Figure 1.1).

(F1) A repeated reading repeats. A second magnet along z reads +1 for all of them.

(F2) Probabilities vary smoothly. A second magnet turned through an angle θ still has two possible readings, and the fraction reading +1 is

$$\cos^2(\theta/2) = \frac{1}{2}(1 + \cos\theta). \tag{1.1}$$

(F3) A reading changes the state. Keep the atoms that read +1 along x ($\theta = \pi/2$), and send them through a magnet along z. Half of them now read −1.

These fractions are the limits that measured frequencies approach.

No arrow with three definite components behaves like this. Its components would not take only two values, and a reading along x could not erase a reading along z. We need a *state* that holds the probabilities for every direction. We need rules that give those probabilities and the state after a reading, and a rule for the motion of a state in a field. Atoms straight from the oven, before any magnet, are taken up in Lecture 3.

Self-check 1.1

1. A second magnet makes the angle $\theta = 2\pi/3$ with the first. What fraction of the kept atoms reads +1?
2. At which angle does every kept atom read −1?

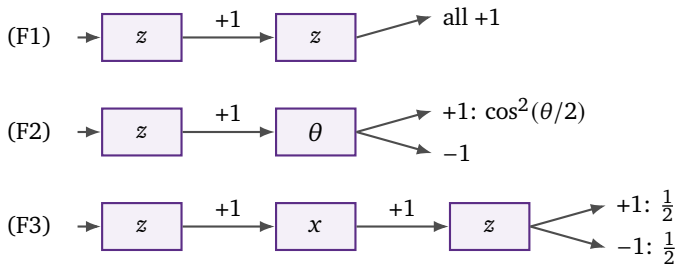


Figure 1.1: The facts (F1)–(F3): Stern–Gerlach magnets in sequence, each labelled by its direction. Between magnets only the beam marked +1 is kept.

1.2 States are rays

The space of the theory is $\mathbb{C}^2$ with the inner product

$$\langle \chi, \psi \rangle = \overline{\chi_1}\,\psi_1 + \overline{\chi_2}\,\psi_2, \tag{1.2}$$

which is antilinear in the first slot (*Linear Algebra for Physics*, from now on *Linear Algebra*, Lecture 7). The standard basis is $|0\rangle = (1,0)$ and $|1\rangle = (0,1)$. In the physicists' notation a vector ψ is also written $|\psi\rangle$, and the inner product $\langle \chi, \psi \rangle$ is written $\langle \chi | \psi \rangle$. For a unit vector ψ, the matrix $|\psi\rangle\langle\psi|$ sends χ to $\langle \psi | \chi \rangle\ \psi$. It is the orthogonal projection onto the line through ψ.

Postulate 1.1 (States). The states of a qubit are described by unit vectors $\psi \in \mathbb{C}^2$. Two unit vectors describe the same state if and only if they differ by a *phase*, that is, $\psi' = e^{i\eta}\psi$ with η real. A state is a *ray* $[\psi] = \{e^{i\eta}\psi : \eta \in \mathbb{R}\}$.

A unit vector is $\psi = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. With $\alpha = a_0 + i a_1$ and $\beta = b_0 + i b_1$ this reads $a_0^2 + a_1^2 + b_0^2 + b_1^2 = 1$, so the unit vectors form the three-sphere S^3 in $\mathbb{R}^4$. Each circle $\{e^{i\eta}\psi\}$ in S^3 is a single state. A phase must therefore be invisible. Every prediction below depends on ψ only through $|\psi\rangle\langle\psi|$, which a phase does not change.

Only the phase in front of the *whole* vector is invisible. The vectors

$$|{+x}\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \quad \text{and} \quad |{-x}\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big)$$

differ by the sign of one component, yet they are orthogonal. Every state is a *superposition* $\alpha\,|0\rangle + \beta\,|1\rangle$ of the two states that the magnet along z reads as $+1$ and -1. A superposition is not a state that is sometimes $|0\rangle$ and sometimes $|1\rangle$. The state $|{+x}\rangle$ reads $+1$ along x with certainty, and neither $|0\rangle$ nor $|1\rangle$ does.

Self-check 1.2

1. Do $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ and $\frac{1}{\sqrt{2}}(i\,|0\rangle - |1\rangle)$ describe the same state?
2. Do $|{+x}\rangle$ and $|{-x}\rangle$?

1.3 The Bloch sphere

The *Pauli matrices* are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are self-adjoint. Multiplying them out gives the *Pauli rule*

$$\sigma_a\sigma_b = \delta_{ab}\mathbb{1} + \mathrm{i}\sum_c \epsilon_{abc}\sigma_c \qquad \text{for } a,b \in \{x,y,z\}, \tag{1.3}$$

with $\epsilon_{xyz} = 1$ and ϵ antisymmetric. Hence $\operatorname{Tr}\sigma_a = 0$ and $\operatorname{Tr}(\sigma_a\sigma_b) = 2\delta_{ab}$. For a vector $\mathbf{v} \in \mathbb{R}^3$ we write $\mathbf{v}\cdot\boldsymbol{\sigma} = v_x\sigma_x + v_y\sigma_y + v_z\sigma_z$.

Definition 1.2 (Bloch vector)**.** The *Bloch vector* of a unit vector $\psi = \alpha|0\rangle + \beta|1\rangle$ is the vector $\mathbf{r}(\psi) \in \mathbb{R}^3$ with components $r_a = \langle\psi, \sigma_a\psi\rangle$. Written out,

$$\mathbf{r}(\psi) = \big(2\operatorname{Re}(\bar\alpha\beta),\ 2\operatorname{Im}(\bar\alpha\beta),\ |\alpha|^2 - |\beta|^2\big). \tag{1.4}$$

Theorem 1.3 (The Bloch sphere)**.** *Let ψ and χ be unit vectors in $\mathbb{C}^2$.*

1. *The Bloch vector is a unit vector, $|\mathbf{r}(\psi)| = 1$, and $\mathbf{r}(e^{\mathrm{i}\eta}\psi) = \mathbf{r}(\psi)$ for every real η.*
2. *With $\mathbf{r} = \mathbf{r}(\psi)$,*
$$|\psi\rangle\langle\psi| = \frac{1}{2}\big(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma}\big). \tag{1.5}$$
3. *Every unit vector $\mathbf{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is the Bloch vector of exactly one state, the ray of*
$$\psi(\theta,\phi) = \cos(\theta/2)\,|0\rangle + e^{\mathrm{i}\phi}\sin(\theta/2)\,|1\rangle\,. \tag{1.6}$$
4. *The overlap of two states is*
$$|\langle\chi,\psi\rangle|^2 = \frac{1}{2}\big(1 + \mathbf{r}(\chi)\cdot\mathbf{r}(\psi)\big). \tag{1.7}$$
So $\chi \perp \psi$ if and only if $\mathbf{r}(\chi) = -\mathbf{r}(\psi)$.

Hence the states of a qubit are the points of the unit sphere, the Bloch sphere.

Proof.

1. By (1.4),
$$|\mathbf{r}|^2 = 4|\alpha|^2|\beta|^2 + \big(|\alpha|^2 - |\beta|^2\big)^2 = \big(|\alpha|^2 + |\beta|^2\big)^2 = 1.$$
A phase multiplies α and β by the same $e^{\mathrm{i}\eta}$, so it leaves $\bar\alpha\beta$, $|\alpha|$ and $|\beta|$ unchanged.
2. We compare entries. The matrix $|\psi\rangle\langle\psi|$ has rows $(|\alpha|^2,\ \alpha\bar\beta)$ and $(\bar\alpha\beta,\ |\beta|^2)$. The right side has rows $\frac{1}{2}(1 + r_z,\ r_x - \mathrm{i}r_y)$ and $\frac{1}{2}(r_x + \mathrm{i}r_y,\ 1 - r_z)$. By (1.4),
$$r_x + \mathrm{i}r_y = 2\bar\alpha\beta, \quad 1 + r_z = 2|\alpha|^2, \quad 1 - r_z = 2|\beta|^2\,.$$
3. Put $\alpha = \cos(\theta/2)$ and $\beta = e^{\mathrm{i}\phi}\sin(\theta/2)$ into (1.4). The half-angle formulas give $\mathbf{r}(\psi(\theta,\phi)) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) = \mathbf{r}$. Next, suppose $\mathbf{r}(\psi) = \mathbf{r}(\chi)$. Then $|\psi\rangle\langle\psi| = |\chi\rangle\langle\chi|$ by (1.5). Applied to ψ, this gives $\psi = \langle\chi|\psi\rangle\,\chi$, a multiple of χ. Both are unit vectors, so the multiple is a phase.
4. With $\mathbf{s} = \mathbf{r}(\chi)$, the projector formula (1.5) and the traces of the Pauli matrices give
$$|\langle\chi|\psi\rangle|^2 = \operatorname{Tr}\big(|\chi\rangle\langle\chi|\ |\psi\rangle\langle\psi|\big) = \frac{1}{4}\operatorname{Tr}\big((\mathbb{1} + \mathbf{s}\cdot\boldsymbol{\sigma})(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})\big) = \frac{1}{2}(1 + \mathbf{s}\cdot\mathbf{r}).$$
This vanishes exactly when $\mathbf{s}\cdot\mathbf{r} = -1$, which for unit vectors means $\mathbf{s} = -\mathbf{r}$. ∎

Remark 1.4 (Antipodal, not perpendicular)**.** Orthogonal states, such as $|0\rangle$ and $|1\rangle$, are *opposite* points of the sphere (Figure 1.2), and $|0\rangle$ and $|+x\rangle$, a quarter turn apart, have overlap $\frac{1}{2}$.

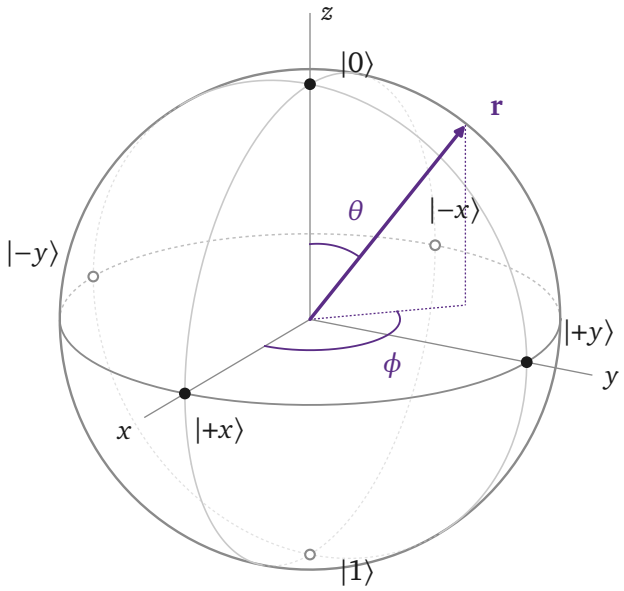


Figure 1.2: The Bloch sphere. The states $|0\rangle$ and $|1\rangle$ are at the poles, and $|\pm x\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ and $|\pm y\rangle = (|0\rangle \pm \mathrm{i}\,|1\rangle)/\sqrt{2}$ on the equator (open markers are behind the sphere). The state (1.6) is at polar angle θ and azimuth ϕ.

Self-check 1.3

1. Find the Bloch vector of $\frac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle)$.
2. Which state is orthogonal to $\psi(\pi/3, 0)$? Write a unit vector for it.
3. Find $|\langle 0|\psi(\pi/3,0)\rangle|^2$.

1.4 Observables and measurement

Postulate 1.5 (Observables). Each measurable quantity of a qubit, an *observable*, is a self-adjoint matrix $A = A^\dagger$ on $\mathbb{C}^2$. Its possible readings are its eigenvalues.

Physicists also say *Hermitian*. This course says self-adjoint. By the spectral theorem (*Linear Algebra*, Thm 10.17), a self-adjoint A is $A = \sum_a a\,E_a$, where a runs over the distinct eigenvalues, E_a is the orthogonal projection onto the eigenspace of a, and $\sum_a E_a = \mathbb{1}$.

Postulate 1.6 (Measurement). An *ideal measurement* of an observable A on the state $[\psi]$ obeys two rules.

1. *Born's rule.* It gives the reading a with probability $\mathrm{Prob}(a) = \langle \psi, E_a\psi\rangle = \|E_a\psi\|^2$.
2. *Lüders' rule.* If the reading is a, which requires $\mathrm{Prob}(a) > 0$, the state immediately afterwards is the ray of $E_a\psi/\|E_a\psi\|$.

Both rules depend only on the ray. They are two separate assumptions, and Lüders' rule does not follow from Born's rule. The change of state is often called *collapse*. The postulate predicts what follows a reading, which (F1) and (F3) require. It claims nothing about how, or whether, a physical process brings the change about. Measurements that are not ideal are studied in *Quantum Mechanics III: Foundations*. A reading of probability 0 does not occur,

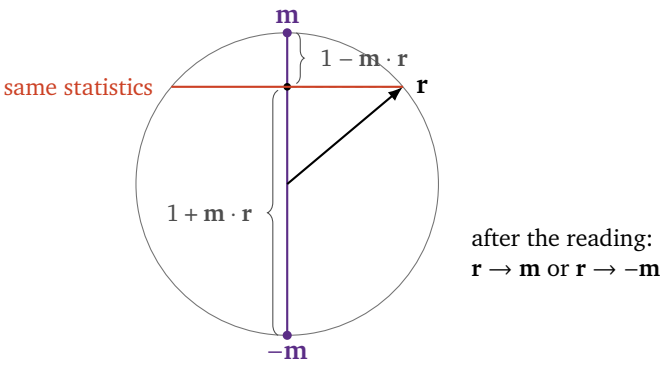


Figure 1.3: Born's rule, drawn in the plane of $\mathbf{m}$ and $\mathbf{r}$. The component $\mathbf{m}\cdot\mathbf{r}$ of $\mathbf{r}$ along $\mathbf{m}$ cuts the diameter into pieces of lengths $1 \pm \mathbf{m}\cdot\mathbf{r}$. Each probability is its piece divided by the diameter 2. The states on the circle of latitude through $\mathbf{r}$ (red, seen edge-on) have the same component, and the same statistics.

and it has no state after it.

Proposition 1.7 (The observables of a qubit). *Every self-adjoint 2×2 matrix is $A = a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$ with $a \in \mathbb{R}$, $b \geq 0$ and $\mathbf{m}$ a unit vector. If $b > 0$, the readings of A are $a \pm b$, with projections*

$$E_\pm = \frac{1}{2}\big(\mathbb{1} \pm \mathbf{m}\cdot\boldsymbol{\sigma}\big), \tag{1.8}$$

and $E_\pm = |\psi_\pm\rangle\langle\psi_\pm|$, where $\psi_\pm$ is the state with Bloch vector $\pm\mathbf{m}$. If $b = 0$, then $A = a\mathbb{1}$ reads a with certainty, and the measurement leaves the state unchanged.

Proof. A self-adjoint matrix has real diagonal entries p, q and off-diagonal entries $u \mp \mathrm{i}v$. It equals $\frac{1}{2}(p+q)\mathbb{1} + u\sigma_x + v\sigma_y + \frac{1}{2}(p-q)\sigma_z$. Put $a = \frac{1}{2}(p+q)$ and $\mathbf{b} = (u, v, \frac{1}{2}(p-q))$.

If $\mathbf{b} = 0$, then $A = a\mathbb{1}$. Its only eigenvalue is a, with $E_a = \mathbb{1}$, so $\mathrm{Prob}(a) = 1$ and Lüders' rule returns the ray of ψ.

If $\mathbf{b} \neq 0$, put $b = |\mathbf{b}|$ and $\mathbf{m} = \mathbf{b}/b$. By the Pauli rule $(\mathbf{m}\cdot\boldsymbol{\sigma})^2 = \mathbb{1}$, so the matrices $E_\pm$ are self-adjoint, with $E_\pm^2 = E_\pm$, $E_+E_- = 0$ and $E_+ + E_- = \mathbb{1}$. Then $A = (a+b)E_+ + (a-b)E_-$ is the spectral decomposition. By the projector formula (1.5), $E_\pm = |\psi_\pm\rangle\langle\psi_\pm|$. ∎

So every observable that is not a multiple of $\mathbb{1}$ is a reading along an axis $\mathbf{m}$, up to the two numbers it reports. A Stern–Gerlach magnet along $\mathbf{m}$ measures $\frac{\hbar}{2}\,\mathbf{m}\cdot\boldsymbol{\sigma}$, where $\hbar$ is Planck's constant divided by 2π. Its readings $\pm\frac{\hbar}{2}$ we have recorded as ± 1.

Corollary 1.8 (Born's rule on the sphere). *In the state with Bloch vector $\mathbf{r}$, a measurement along a unit vector $\mathbf{m}$ reads ± 1 with probabilities*

$$\mathrm{Prob}(\pm 1) = \frac{1}{2}\big(1 \pm \mathbf{m}\cdot\mathbf{r}\big). \tag{1.9}$$

After the reading ± 1, the Bloch vector is $\pm\mathbf{m}$.

Proof. By the overlap formula (1.7), $\langle\psi, E_\pm\psi\rangle = |\langle\psi_\pm|\psi\rangle|^2 = \frac{1}{2}(1 \pm \mathbf{m}\cdot\mathbf{r})$. The vector $E_\pm\psi = \langle\psi_\pm|\psi\rangle\,\psi_\pm$ is a multiple of $\psi_\pm$. When the reading occurs it is nonzero, and its normalisation lies on the ray of $\psi_\pm$. ∎

Figure 1.3 draws (1.9). A measurement along $\mathbf{m}$ sees only the component $\mathbf{m}\cdot\mathbf{r}$ of the state along its axis, the shadow of $\mathbf{r}$ on the axis, as it were. The facts of § 1.1 follow. After

the reading +1 along z, the Bloch vector is $\mathbf{e}_z$, and a second reading along z is certain (F1). An axis at angle θ to $\mathbf{e}_z$ has $\mathbf{m}\cdot\mathbf{r}=\cos\theta$, which gives (1.1) (F2). After the reading +1 along x, the Bloch vector is $\mathbf{e}_x$, whose component along z is 0 (F3). The reading along x did not reveal a hidden value along z. It replaced the state.

Remark 1.9 (One observable sees one component)**.** All the states on one circle of latitude about $\mathbf{m}$, that is, all $\mathbf{r}$ with the same $\mathbf{m}\cdot\mathbf{r}$, give the same statistics for every observable $a\mathbb{1}+b\,\mathbf{m}\cdot\boldsymbol{\sigma}$. Unless the circle is a single pole, a single observable therefore cannot determine the state. Three observables along independent axes can, and Exercise Sheet 1 shows how.

Self-check 1.4

1. Find the readings and the projections of $A=\mathbb{1}+2\sigma_x$.
2. In the state with $\mathbf{r}=(\frac{\sqrt{3}}{2},0,\frac{1}{2})$, a measurement along $\mathbf{m}=(\mathbf{e}_x+\mathbf{e}_z)/\sqrt{2}$ reads +1 with what probability, and what is the Bloch vector afterwards?

1.5 The Hamiltonian and time evolution

Postulate 1.10 (Evolution)**.** Between measurements, the state vector obeys the *Schrödinger equation*

$$\mathrm{i}\hbar\,\frac{\mathrm{d}\psi_t}{\mathrm{d}t}=H\psi_t. \tag{1.10}$$

Here the *Hamiltonian* H is the observable whose readings are the energies.

Lemma 1.11 (Evolution is unitary)**.** *The only solution with $\psi_0=\psi$ is $\psi_t=U_t\psi$, where $U_t=e^{-\mathrm{i}tH/\hbar}$. For every observable A,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\psi_t,A\psi_t\rangle=\frac{\mathrm{i}}{\hbar}\langle\psi_t,[H,A]\,\psi_t\rangle,\qquad \text{where } [H,A]=HA-AH. \tag{1.11}$$

In particular, U_t is unitary and maps rays to rays.

Proof. Existence and uniqueness are *Linear Algebra*, Prop. 6.21, for the matrix $-\mathrm{i}H/\hbar$. Next we differentiate the inner product. A constant leaves the first slot conjugated, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\psi_t,A\psi_t\rangle=\frac{\mathrm{i}}{\hbar}\langle H\psi_t,A\psi_t\rangle-\frac{\mathrm{i}}{\hbar}\langle\psi_t,AH\psi_t\rangle.$$

Since $H=H^\dagger$, the first term is $\frac{\mathrm{i}}{\hbar}\langle\psi_t,HA\psi_t\rangle$, which gives (1.11). For $A=\mathbb{1}$ the commutator vanishes, so $\|\psi_t\|$ is constant for every initial vector. A matrix that preserves every norm is unitary (*Linear Algebra*, Thm 9.21). Finally, $U_t(e^{\mathrm{i}\eta}\psi)=e^{\mathrm{i}\eta}U_t\psi$. ∎

So H must be self-adjoint, or the total probability $\|\psi_t\|^2$ would drift. By Proposition 1.7, every Hamiltonian of a qubit is either $c\mathbb{1}$, which moves no state, or

$$H=c\mathbb{1}+\frac{\hbar\nu}{2}\,\mathbf{n}\cdot\boldsymbol{\sigma}\qquad\text{with } \nu>0 \tag{1.12}$$

and $\mathbf{n}$ a unit vector. Its energies are $E_\pm=c\pm\frac{1}{2}\hbar\nu$, so $\nu=(E_+-E_-)/\hbar$.

Theorem 1.12 (Precession)**.** *Under the Hamiltonian (1.12), the Bloch vector of ψ_t obeys*

$$\dot{\mathbf{r}}=\nu\,\mathbf{n}\times\mathbf{r}. \tag{1.13}$$

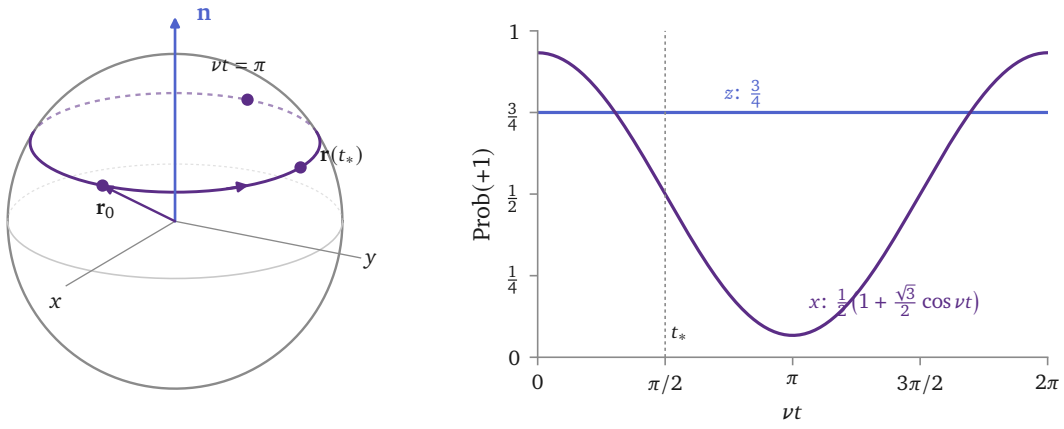


Figure 1.4: Example 1.13. Left: the Bloch vector turns about $\mathbf{n} = \mathbf{e}_z$ on the circle $r_z = \frac{1}{2}$, and reaches $\mathbf{r}(t_*)$ after a quarter turn. Right: Prob(+1) along z and along x, against νt.

Hence

$$\mathbf{r}(t) = \cos(\nu t)\,\mathbf{r}_0 + \sin(\nu t)\,\mathbf{n}\times\mathbf{r}_0 + (1-\cos\nu t)(\mathbf{n}\cdot\mathbf{r}_0)\,\mathbf{n}.$$

The sphere turns rigidly about $\mathbf{n}$ *through the angle* νt*, anticlockwise when seen from the tip of* $\mathbf{n}$.

Proof. Put $A = \sigma_a$ in (1.11). The term $c\mathbb{1}$ commutes with σ_a. By the Pauli rule, $[\sigma_b, \sigma_a] = 2\mathrm{i}\sum_c \epsilon_{bac}\sigma_c$, so

$$\frac{\mathrm{i}}{\hbar}[H,\sigma_a] = \frac{\mathrm{i}\nu}{2}\sum_b n_b\,[\sigma_b,\sigma_a] = \nu\sum_{b,c}\epsilon_{abc}\,n_b\sigma_c.$$

Taking $\langle\psi_t,\,\cdot\,\psi_t\rangle$ of both sides gives $\dot r_a = \nu\sum_{b,c}\epsilon_{abc}n_b r_c = \nu(\mathbf{n}\times\mathbf{r})_a$, which is (1.13). This linear equation has exactly one solution with $\mathbf{r}(0) = \mathbf{r}_0$ (*Linear Algebra*, Prop. 6.21). The displayed rotation equals $\mathbf{r}_0$ at $t = 0$, and differentiating it with $\mathbf{n}\times(\mathbf{n}\times\mathbf{r}_0) = (\mathbf{n}\cdot\mathbf{r}_0)\mathbf{n} - \mathbf{r}_0$ gives $\nu\,\mathbf{n}\times\mathbf{r}(t)$. ∎

The eigenstates of H, at $\mathbf{r} = \pm\mathbf{n}$, do not move. They are the *stationary states*. The component $\mathbf{n}\cdot\mathbf{r}$ is constant, so the energy statistics $\frac{1}{2}(1\pm\mathbf{n}\cdot\mathbf{r})$ of every state are constant, even when the state moves.

A spin in a nonzero uniform magnetic field $\mathbf{B}$ has $H = -\gamma\,\frac{\hbar}{2}\boldsymbol{\sigma}\cdot\mathbf{B}$, with γ the magnetic moment per unit spin (the *gyromagnetic ratio*). For $\gamma > 0$ this is (1.12) with $c = 0$, $\mathbf{n} = -\mathbf{B}/|\mathbf{B}|$ and the *Larmor frequency* $\nu = \gamma\,|\mathbf{B}|$.

Self-check 1.5

1. Under $H = \frac{\hbar\nu}{2}\sigma_x$, the state $|0\rangle$ evolves for the time $\pi/(2\nu)$. Where is its Bloch vector then?
2. Which states does $H = c\mathbb{1} + \frac{\hbar\nu}{2}\mathbf{n}\cdot\boldsymbol{\sigma}$ leave where they are?

1.6 A worked sequence

Example 1.13 (Evolve, then measure twice)**.** Figure 1.4 shows this example.

1. *Prepare.* The state is

$$\psi_0 = \tfrac{\sqrt{3}}{2}\,|0\rangle + \tfrac{1}{2}\,|1\rangle, \qquad \mathbf{r}_0 = \left(\tfrac{\sqrt{3}}{2}, 0, \tfrac{1}{2}\right),$$

which is (1.6) with $\theta = \pi/3$ and $\phi = 0$.

2. *Evolve.* The Hamiltonian and the time are

$$H = \tfrac{\hbar\nu}{2}\sigma_z \quad \text{and} \quad t_* = \pi/(2\nu),$$

a quarter turn. Theorem 1.12 turns the Bloch vector about z to $\mathbf{r}(t_*) = \left(0, \tfrac{\sqrt{3}}{2}, \tfrac{1}{2}\right)$.

3. *Read the statistics.* Along z, $\mathrm{Prob}(+1) = \tfrac{3}{4}$ and $\mathrm{Prob}(-1) = \tfrac{1}{4}$ at every time. Along x,

$$\mathrm{Prob}(+1) = \frac{1}{2}\left(1 + \frac{\sqrt{3}}{2}\cos(\nu t)\right).$$

This is $\frac{2+\sqrt{3}}{4} \approx 0.933$ at $t = 0$, with $\frac{2-\sqrt{3}}{4} \approx 0.067$ for -1, and it is $\frac{1}{2}$ at t_*. The relative phase of $|0\rangle$ and $|1\rangle$ changes, and the reading along x sees it.

4. *Measure x, then z at once.* At t_*, measure along x and suppose the reading is $+1$, which has probability $\frac{1}{2}$. The state is now $|+x\rangle$. A second reading along x, made at once, is $+1$ with certainty. A reading along z, made at once, gives ± 1 with probability $\frac{1}{2}$ each. So "$+1$ along x, then $+1$ along z" has probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Without the reading along x, $+1$ along z had probability $\frac{3}{4}$.

Self-check 1.6

1. In Example 1.13, when is the reading $+1$ along x first least likely, and how likely is it then?
2. If the reading along x at t_* is -1 instead, what is the probability of $+1$ along z at once?

1.7 Is the Bloch sphere a phase space?

In classical mechanics the state of a system is a point (q, p) of its *phase space*, and the energy, a function $h(q, p)$, moves the points by *Hamilton's equations* $\dot{q} = \partial h/\partial p$ and $\dot{p} = -\partial h/\partial q$. Take $\mathbf{n} = \mathbf{e}_z$. Away from the poles, use the local coordinates $q = \phi$ and $p = \frac{\hbar}{2}\cos\theta$ on the sphere, and the energy function $h = c + \frac{\hbar\nu}{2}\cos\theta = c + \nu p$. Then $\dot{q} = \nu$ and $\dot{p} = 0$: the azimuth grows at the rate ν at a fixed height, which is the precession about $\mathbf{e}_z$. The poles, where the coordinates fail, are the stationary states. Exercise Sheet 1 does the same for every axis. For the *motion*, then, the sphere is a phase space.

For the *readings*, it is not. On a classical phase space a point fixes the value of every observable. A point of the Bloch sphere fixes the reading along $\mathbf{m}$ only when $\mathbf{m} = \pm\mathbf{r}$, and along any other axis only its probabilities. A quantum state moves like a point of a phase space, but it holds probabilities, not values.

Self-check 1.7

1. Along which axes does the state $|+x\rangle$ give a certain reading?
2. With $h = c + \nu p$, Hamilton's equations give $\dot{p} = 0$. What does this say about the energy readings?

The qubit, as a quantum system

States: the rays of $\mathbb{C}^2$, which are the points $\mathbf{r}$ of the Bloch sphere (Postulate 1.1 and Theorem 1.3). *Observables:* the self-adjoint matrices $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$. Unless $b = 0$, each is a reading along an axis $\mathbf{m}$ (Postulate 1.5 and Proposition 1.7). *Evolution:* $\psi_t = e^{-\mathrm{i}tH/\hbar}\psi$. The Hamiltonian $H = c\mathbb{1} + \frac{\hbar\nu}{2}\mathbf{n}\cdot\boldsymbol{\sigma}$ turns the sphere about $\mathbf{n}$ at the rate ν (Postulate 1.10 and Theorem 1.12). *Measurement:* along $\mathbf{m}$, the reading

is ± 1 with probability $\frac{1}{2}(1 \pm \mathbf{m} \cdot \mathbf{r})$, and the state afterwards is $\pm\mathbf{m}$ (Postulate 1.6 and Corollary 1.8).

Summary.

- *The Bloch sphere*: states are points, and orthogonal states are antipodal (Theorem 1.3).
- *Born's rule on the sphere* and *precession*: the statistics of a measurement depend on one component, and a Hamiltonian turns the sphere (Corollary 1.8 and Theorem 1.12). A point of the sphere holds probabilities, not values.

Lecture 2 drives the qubit with a field that turns in time, and Lecture 3 fills the inside of the sphere.

Self-check answers.

1.1 1. $\cos^2(\pi/3) = \frac{1}{4}$. 2. $\theta = \pi$, the magnet turned over.

1.2 1. Yes: the second is i times the first. 2. No: they are orthogonal. A sign on one component is not a phase.

1.3 1. $(0, 1, 0)$, the state $|{+y}\rangle$. 2. The ray of $\frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle$, at the antipode $\psi(2\pi/3, \pi)$. 3. $\frac{1}{2}(1 + \cos\frac{\pi}{3}) = \frac{3}{4}$.

1.4 1. This is $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$ with $a = 1$, $b = 2$, $\mathbf{m} = \mathbf{e}_x$: readings 3 and -1, projections $\frac{1}{2}(\mathbb{1} \pm \sigma_x)$. 2. $\mathbf{m}\cdot\mathbf{r} = (\sqrt{6} + \sqrt{2})/4$, so $\mathrm{Prob}(+1) = (4 + \sqrt{6} + \sqrt{2})/8 \approx 0.983$, and afterwards $\mathbf{r} = \mathbf{m}$.

1.5 1. A quarter turn about $\mathbf{e}_x$ takes $\mathbf{e}_z$ to $-\mathbf{e}_y$: the state $|{-y}\rangle$. 2. The two at $\mathbf{r} = \pm\mathbf{n}$, the eigenstates of H.

1.6 1. At $t = \pi/\nu$, half a turn, with probability $\frac{2-\sqrt{3}}{4} \approx 0.067$. 2. $\frac{1}{2}$: the state is $|{-x}\rangle$.

1.7 1. Along $\pm\mathbf{e}_x$ only: $+1$ along $\mathbf{e}_x$, -1 along $-\mathbf{e}_x$. 2. The statistics of the energy do not change in time.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §1.1, pp. 1–5. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Ch. IV, §§B-1-c to B-3-b, pp. 403–411. Nielsen and Chuang, *Quantum Computation and Quantum Information*, §1.2, pp. 13–16, and §2.2.5, pp. 87–88. Heinosaari and Ziman, *The Mathematical Language of Quantum Theory*, Ex. 2.21, pp. 62–63. *Linear Algebra for Physics*, Prop. 6.21, Thm 9.21, Thm 10.17.